

Air Ticket Reservation System

A Web-Based PHP & MySQL Project

Submitted By: | SEAT NO:

Guide:

College: S.M. Chaudhari College, Jalgaon

Academic Year: 2025–26 | BCA-3rd

Tech Stack

PHP 8.x

MySQL 8.x

HTML5 / CSS3

JavaScript

Apache (XAMPP)

Introduction

What is this project?

The Air Ticket Reservation System is a full-stack web application that enables passengers to search flights, book seats, manage passenger details, and confirm payments online — while giving administrators complete control over flight inventory, delays, and bookings.



Problem

Manual paper-based ticketing is slow, error-prone, and inaccessible remotely. No real-time seat or delay updates.



Solution

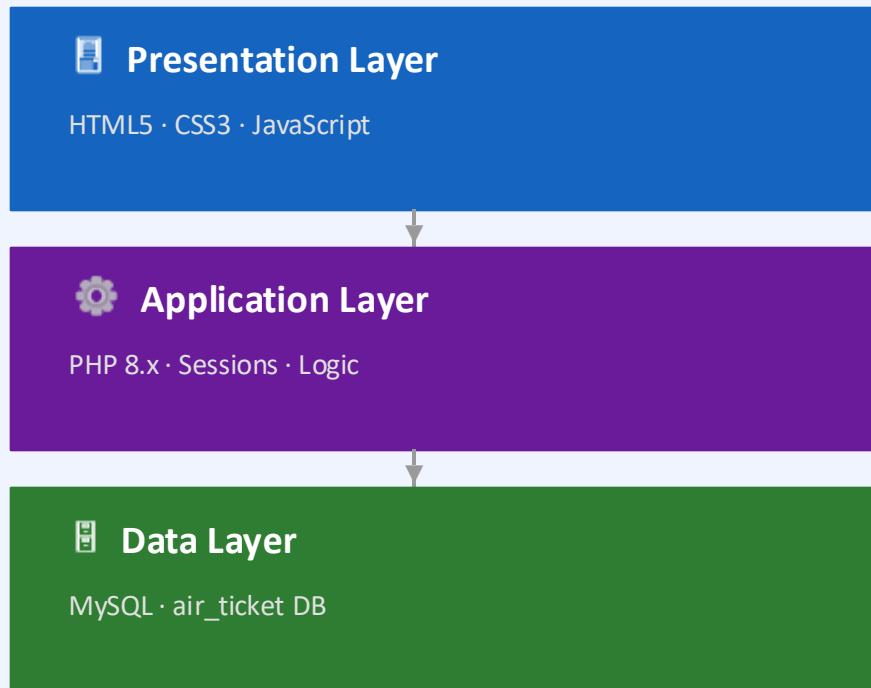
A PHP + MySQL web app that allows 24/7 online booking, live flight status, and admin-controlled seat management.



Outcome

Paperless booking, real-time seat inventory, delay management, OTP-based security, and paginated booking history.

System Architecture



Key Modules



User Module

Register · Login · OTP · Search · Book · History



Admin Module

Add Flight · View Flights · Edit · View Bookings



Database

3 tables: users · flights · bookings



Security

Session-based auth for both roles

Database Design

users

id (PK)
name
email
password

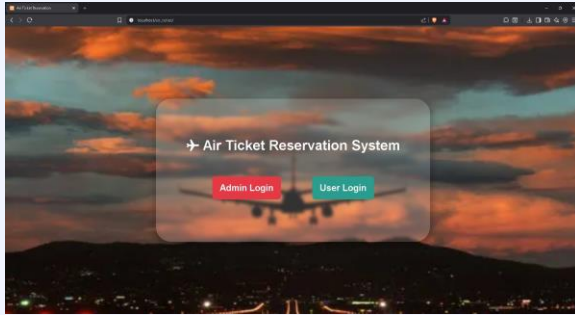
flights

id (PK)
flight_no
flight_name
source
destination
flight_date
departure_time
arrival_time
seats
price
status
delay_minutes

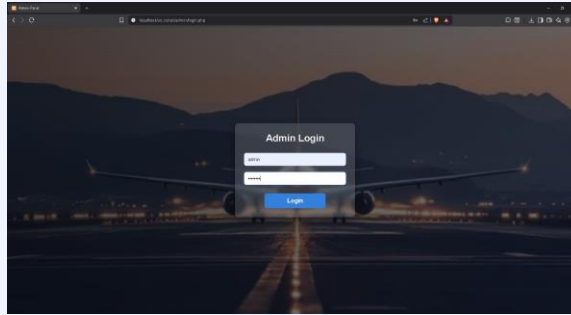
bookings

id (PK)
flight_no (FK)
email
passenger_names
passenger_ages
seats
mobile
payment_Status
booking_date

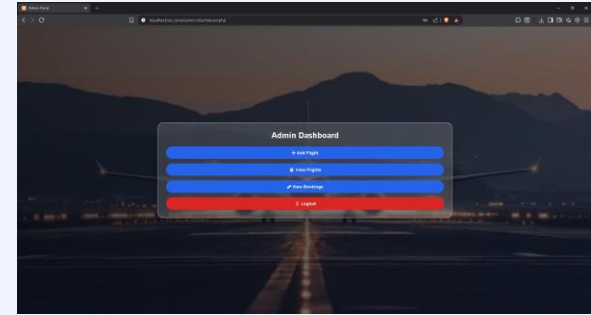
Screenshots – Entry Points



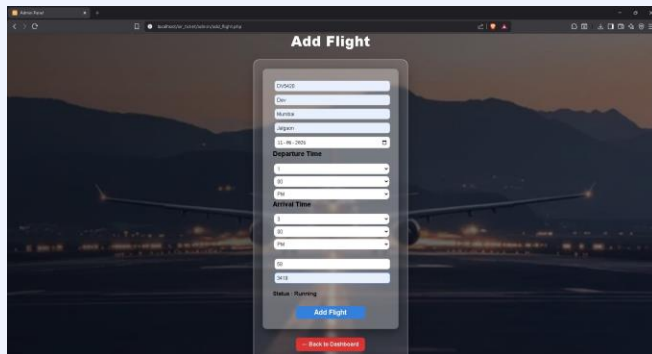
Home Page



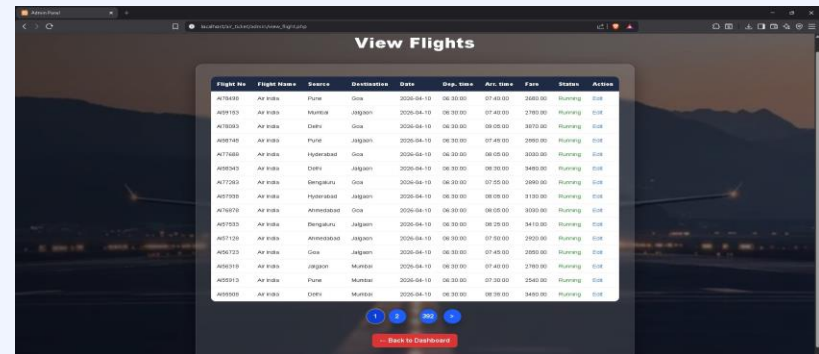
Admin Login



Admin Dashboard

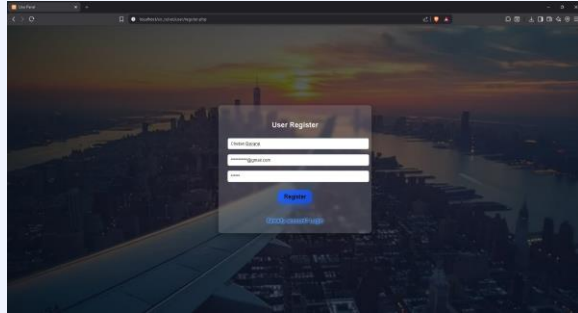


Add Flight

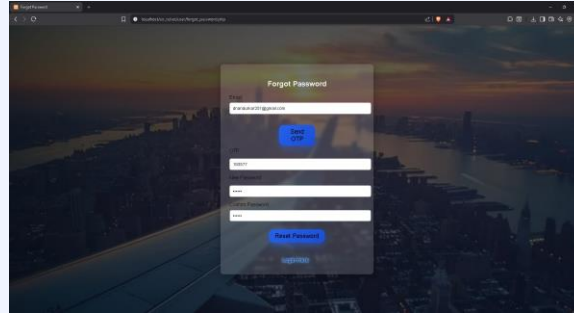


View Flights (Admin)

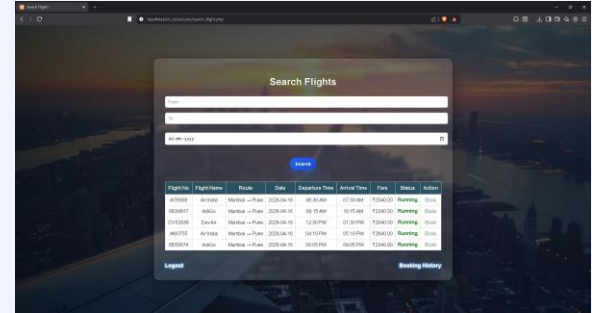
Screenshots – User Booking Flow



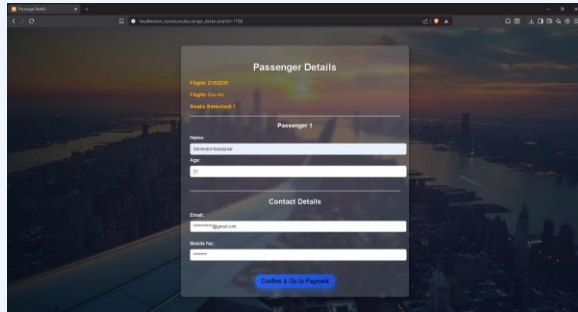
1. Register



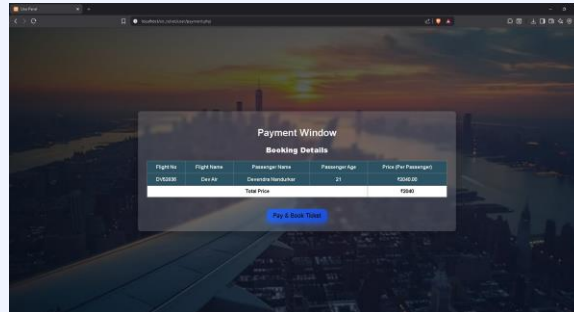
2. Forgot Password



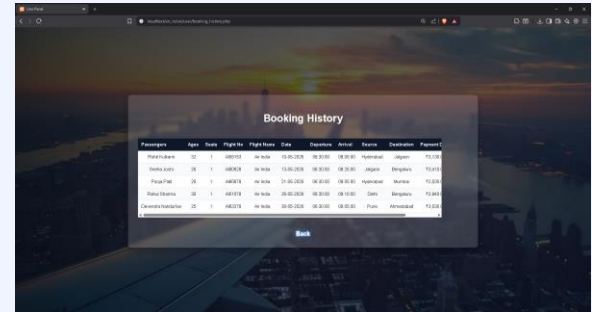
3. Search Flights



4. Passenger Details



5. Payment



6. Booking History

Key Features



Session Auth

Separate admin & user sessions



Seat Inventory

Real-time seat count control



City Autocomplete

JS-powered city suggestions



Booking History

Per-user email-linked records



Delay Management

Auto-recalculates revised times



OTP Password

6-digit OTP-based recovery



Auto-Cancellation

Delay > 300 min = auto cancel



Pagination

15/10 rows — all list pages

Testing Summary

Black-Box Testing

Tested user-facing features without code knowledge. All form inputs, validations, and redirects verified.



White-Box Testing

Verified internal logic: delay calc, auto-cancel (>300 min), seat decrement, session handling.



Unit Testing

Individual PHP functions and SQL queries tested in isolation with test data.



Integration Testing

End-to-end flow: Search → Book → Payment → History. Session vars checked at each step.



Future Scope

Password Hashing

bcrypt via password_hash()

Real Email OTP

PHPMailer / SMTP integration

SQL Injection Fix

Migrate all queries to prepared statements

Live Hosting

Deploy on cloud/cPanel server

Payment Gateway

Razorpay / PayU integration

Mobile App

REST API + Android/iOS app



Thank You

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Guide:

Technology Stack: PHP · MySQL · HTML5 · CSS3 · JavaScript
